

St George Girls High School

Year 12

Common Test 3

June 2009



Mathematics

Extension 2

General Instructions

- Working time – 75 minutes
- Reading time – 5 minutes
- Write using blue or black pen.
- Board-approved calculators may be used.
- All necessary working must be shown in every question.
- Write on one side of the page only.
- Start each question on a new page.

Total marks – 52

- Attempt Questions 1 – 3
- Questions are not all of equal value.

Students are advised that this is a School Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (18 marks) – Start a New Page

Marks

- a) Find $\int \frac{\sin x}{\cos^4 x} dx$ 2
- b) Use completion of the square to find $\int \frac{1}{\sqrt{5+4x-x^2}} dx$ 2
- c) Using the substitution $t = \tan \frac{\theta}{2}$ (ie $\theta = 2\tan^{-1}t$) find the value of $\int_0^{\frac{\pi}{2}} \frac{1}{2 + 3 \sin \theta + 2 \cos \theta} d\theta$ 4
- d) (i) Show that $\sqrt{x^2 + 16} - \frac{16}{\sqrt{x^2+16}} = \frac{x^2}{\sqrt{x^2+16}}$ 1
- (ii) Use integration by parts to find the value of $\int_0^3 \sqrt{x^2 + 16} dx$ 4
- e) $I_n = \int_0^{\frac{\pi}{2}} \cos^n x dx$
- (i) Show that $I_n = \frac{n-1}{n} I_{n-2}$ 3
- (ii) Hence find the value of $\int_0^{\frac{\pi}{2}} \cos^5 x dx$ 2

Question 2 – (17 marks) – Start a New Page

Marks

a) Given that $\alpha + \beta + \gamma = -3$
 $\alpha^2 + \beta^2 + \gamma^2 = 29$
 $\alpha\beta\gamma = -6$

form the monic cubic equation whose roots are α, β and γ .

3

b) α, β, γ are the roots of the polynomial equation $3x^3 - 4x^2 + 5x + 6 = 0$

Form a polynomial equation with integer coefficients and with roots

(i) $\alpha + 3, \beta + 3, \gamma + 3$

2

(ii) $\alpha^2, \beta^2, \gamma^2$

2

c) (i) Show that if a polynomial equation $P(x) = 0$ has a root $x = \alpha$ of multiplicity n , then the polynomial equation $P'(x) = 0$ has $x = \alpha$ as a root of multiplicity $(n - 1)$.

3

(ii) Show that the cubic equation $ax^3 + bx^2 + c = 0$ has a double root when $27a^2c + 4b^3 = 0$ where a, b, c are all non-zero.

3

d) (i) Find real numbers A, B and C so that

$$\frac{5x+2}{(x^2+4)(x+2)} \equiv \frac{Ax+B}{x^2+4} + \frac{C}{x+2} \text{ for all } x \neq -2$$

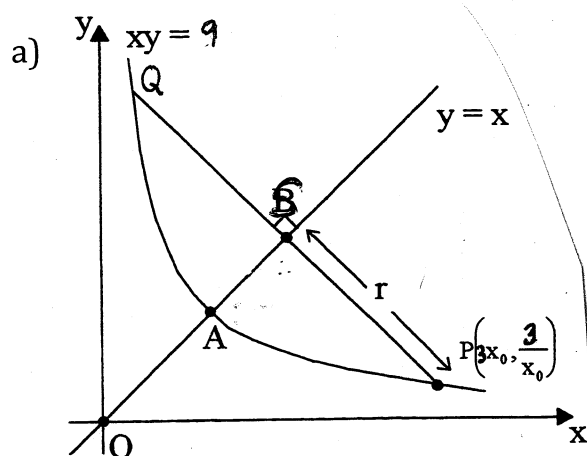
2

(ii) Hence find $\int \frac{5x+2}{(x^2+4)(x+2)} dx$

2

Question 3 - (17 marks) - Start a New Page

Marks



The diagram shows the function $xy = 9$, for $x > 0$

- (i) Write the coordinates of A and find the length of OA .

Given that the eccentricity of the hyperbola is $\sqrt{2}$, find the length of OS , where S is the focus.

2

- (ii) Using the result that $y = x$ is an axis of symmetry of $xy = 9$, and given that P has coordinates $\left(3x_0, \frac{3}{x_0}\right)$ write coordinates for Q in terms of x_0 .

Hence show that $2r^2 = \left(3x_0 - \frac{3}{x_0}\right)^2$

2

- (iii) Using Part (ii), show that $OP^2 = 2(9 + r^2)$

2

- (iv) Using Part (iii), or otherwise, find the length of the latus rectum.

2

- b) $P\left(5p, \frac{5}{p}\right)$ and $Q\left(5q, \frac{5}{q}\right)$ are points on the same branch of the hyperbola $xy = 25$.

- (i) Show that the equation of the tangent at P is $x + p^2y = 10p$ and hence find the point of intersection, T , of the tangents at P and Q .

4

- (ii) Show that the equation of the chord PQ is $x + pqy = 5(p + q)$

2

- (iii) If the chord PQ passes through the point $(0, -2)$ show that $\frac{p+q}{pq} = -\frac{2}{5}$.
Hence find the equation of the locus of T as P and Q vary under this condition.

3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1

$$\begin{aligned} \textcircled{a} \int \cos^4 x \cdot \sin x \, dx &= - \int (\cos x)^{-4} \cdot -\sin x \, dx \\ &= - \frac{(\cos x)^{-3}}{-3} + C \\ &= \frac{1}{3 \cos^3 x} + C \end{aligned}$$

$$\begin{aligned} \textcircled{b} \int \frac{dx}{\sqrt{5-(x^2-4x)}} &= \int \frac{dx}{\sqrt{5-(x^2-4x+4)+4}} \\ &= \int \frac{dx}{\sqrt{9-(x-2)^2}} \\ &= \sin^{-1} \left(\frac{x-2}{3} \right) + C \quad \left[-\cos^{-1} \left(\frac{2-x}{3} \right) \right] \end{aligned}$$

③ $t = \tan \frac{\theta}{2}$

$\therefore 2 \tan^{-1} t = \theta$

$\frac{2}{1+t^2} \cdot dt = d\theta$

When $\theta = 0$, $t = 0$

$\theta = \frac{\pi}{2}$, $t = 1$

$\sin \theta = \frac{2t}{1+t^2}$

$\cos \theta = \frac{1-t^2}{1+t^2}$

Substitution gives

$$\int_0^1 \frac{1}{2 + \frac{6t}{1+t^2} + \frac{2-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{1+t^2}{2+2t^2+6t+2-t^2} \cdot \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{dt}{2+3t}$$

$$= \frac{1}{3} \ln |2+3t| \Big|_0^1$$

$$= \frac{1}{3} (\ln 5 - \ln 2)$$

$$= \frac{1}{3} \ln \left(\frac{5}{2} \right)$$

$$\begin{aligned} \textcircled{d} \text{ (i)} \quad \frac{\sqrt{x^2+16}}{\sqrt{x^2+16}} - \frac{16}{\sqrt{x^2+16}} &= \frac{(x^2+16)-16}{\sqrt{x^2+16}} \\ &= \frac{x^2}{\sqrt{x^2+16}} \end{aligned}$$

$$\textcircled{ii} \int_0^3 \sqrt{x^2+16} \, dx = \sqrt{x^2+16} \cdot x \Big|_0^3 - \int_0^3 \frac{x \cdot x}{\sqrt{x^2+16}} \, dx$$

$$\left[\begin{array}{l} \text{let } u = \sqrt{x^2+16} \quad \text{and } \frac{du}{dx} = 1 \\ \frac{du}{dx} = \frac{x}{\sqrt{x^2+16}} \quad v = x \end{array} \right] = 15 - \int_0^3 \frac{x^2}{x^2+16} \, dx$$

$$\text{In (i)} \int_0^3 \sqrt{x^2+16} \, dx = 15 - \left[\int_0^3 \sqrt{x^2+16} \, dx - \int_0^3 \frac{16}{\sqrt{x^2+16}} \, dx \right]$$

$$\therefore 2 \int_0^3 \sqrt{x^2+16} \, dx = 15 + 16 \int_0^3 \frac{dx}{\sqrt{x^2+16}}$$

$$= 15 + 16 \left[\ln (x + \sqrt{x^2+16}) \right]_0^3$$

$$= 15 + 16 [\ln 8 - \ln 4]$$

$$= 15 + 16 \ln 2$$

$$\therefore \int_0^3 \sqrt{x^2+16} \, dx = \frac{1}{2} (15 + 16 \ln 2)$$

$$e) I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$$

(i) Then

$$I_n = \int_0^{\frac{\pi}{2}} \cos^{n-1} x \cdot \cos x \, dx$$

$$\text{Let } u = \cos^{n-1} x \quad \text{and } dv = \cos x$$

$$\frac{du}{dx} = (n-1) \cos^{n-2} x \cdot (-\sin x) \quad v = \sin x$$

$$\therefore I_n = \left[\cos^{n-1} x \cdot \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \cdot (n-1) \cos^{n-2} x \cdot (-\sin x) \, dx$$

$$= 0 + (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x (\sin^2 x) \, dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x (1 - \cos^2 x) \, dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x \, dx - (n-1) \int_0^{\frac{\pi}{2}} \cos^n x \, dx$$

$$I_n = (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x \, dx - (n-1) I_n$$

$$\text{So } (n-1) I_n + I_n = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$I_n = \frac{(n-1)}{n} I_{n-2}$$

$$(ii) I_5 = \frac{4}{5} I_3$$

$$\text{So } I_3 = \frac{2}{3} I_1$$

$$I_3 = \frac{2}{3} I_1$$

$$I_1 = \int_0^{\frac{\pi}{2}} \cos x \, dx$$

$$= [\sin x]_0^{\frac{\pi}{2}}$$

$$= 1$$

$$I_3 = \frac{4}{3} \times I_1$$

$$= \frac{4}{3} \times 1$$

$$I_5 = \frac{8}{15}$$

Q2

@ If α, β, γ are roots of cubic

$$\text{the } P(x) = (x-\alpha)(x-\beta)(x-\gamma) \\ = x^3 - (\alpha+\beta+\gamma)x^2 + (\alpha\beta+\alpha\gamma+\beta\gamma)x - \alpha\beta\gamma$$

$$\text{We } (\alpha+\beta+\gamma)^2 = 2(\alpha\beta+\alpha\gamma+\beta\gamma) = \alpha^2 + \beta^2 + \gamma^2$$

$$\therefore 2(\alpha\beta+\alpha\gamma+\beta\gamma) = (-3)^2 - 29 \\ = -20$$

$$\alpha\beta+\alpha\gamma+\beta\gamma = -10$$

$$\text{So } P(x) = x^3 + 3x^2 - 10x + 6$$

(b) (i) If $x = \alpha + \beta$
then $x - \beta = \alpha$

Since α is a root, then

$$3(x-\beta)^3 - 4(x-\beta)^2 + 5(x-\beta) + 6 = 0 \text{ is true.}$$

$$3[x^3 - 9x^2\beta + 27x\beta^2 - 27\beta^3] - 4[x^2 - 2x\beta + \beta^2] + 5x - 5\beta + 6 = 0$$

$$3x^3 - 27x^2\beta + 81x\beta^2 - 81\beta^3 - 4x^2 + 8x\beta - 4\beta^2 + 5x - 5\beta + 6 = 0$$

$$3x^3 - 31x^2\beta + 110x\beta^2 - 128\beta^3 = 0$$

(ii) If $x = \alpha^2$

then $\pm\sqrt{x} = \alpha$

Since α is a root, then

$$3(\pm\sqrt{x})^3 - 4(\pm\sqrt{x})^2 + 5(\pm\sqrt{x}) + 6 = 0$$

$$3x\sqrt{x} - 4x + 5\sqrt{x} + 6 = 0$$

$$\sqrt{x} (3x+5) = (4x-6)$$

then

Square both sides $x(9x^2 + 30x + 25) = 16x^2 + 48x + 36$

$$9x^3 + 30x^2 + 25x = 16x^2 + 48x + 36$$

$$9x^3 + 14x^2 + 73x - 36 = 0$$

[same for $-5x$]

(c) (i) let $P(x) = (x - \alpha)^n Q(x)$

α is root of multiplicity n of $P(x)$ and is not a root of polynomial $Q(x)$ i.e. $Q(\alpha) \neq 0$

$$\text{then } P'(x) = n(x - \alpha)^{n-1} Q(x) + (x - \alpha)^n Q'(x)$$

$$= (x - \alpha)^{n-1} [nQ(x) + (x - \alpha)Q'(x)]$$

$$= (x - \alpha)^{n-1} R(x), \quad R(x) \in \text{polynomial}$$

$$R(x) = nQ(x) + (x - \alpha)Q'(x) \quad \text{such that } R(\alpha) \neq 0$$

$$\therefore nQ(\alpha) \neq 0 \text{ since } n > 1 \text{ and } Q(\alpha) \neq 0$$

Then α is root of multiplicity $(n-1)$ of $P'(x)$.

(ii) $P(x) = ax^3 + bx^2 + c$

$$P'(x) = 3ax^2 + 2bx$$

If double root then $x[3ax + 2b] = 0$

Now $x=0$ not a root of $P(x)$ since c is non-zero!

$$\therefore x = -\frac{2b}{3a} \quad \text{i.e.}$$

a root of $P(x)$

$$\text{i.e. } a\left(-\frac{2b}{3a}\right)^3 + b\left(-\frac{2b}{3a}\right)^2 + c = 0$$

$$\frac{-8b^3}{27a^2} + \frac{4b^3}{9a^2} + c = 0$$

$$-8b^3 + 12b^3 + 27ac = 0 \quad \left[\text{multiply by } 27a^2 \right. \\ \left. \text{since a non-zero} \right]$$

$$27a^2c + 4b^3 = 0$$

Required condition for double root.

(d) (i) $\frac{5x+2}{(x^2+4)(x+2)} = \frac{Ax+B}{x^2+4} + \frac{C}{x+2} \quad (x \neq -2)$

$$\begin{aligned} \text{the } 5x+2 &= (Ax+B)(x+2) + C(x^2+4) \\ &= Ax^2 + 2Ax + Bx + 2B + Cx^2 + 4C \\ &= (A+C)x^2 + (2A+B)x + 2B+4C \end{aligned}$$

Equating coeff. (I) $A+C=0 \Rightarrow A=-C$

(II) $2A+B=5$

(III) $2B+4C=2 \Rightarrow B+2C=1$

Sub I in II $B+2(-A)=1 \Rightarrow -2A+B=1$

the II + III $\begin{matrix} 2A+B=5 \\ -2A+B=1 \end{matrix} \Rightarrow 2B=6$
 $B=3$

Sub $2A+3=5 \quad \{ \quad 3+2C=1$
 $A=1 \quad \quad \quad C=-1$

$$\therefore \frac{5x+2}{(x^2+4)(x+2)} = \frac{x+3}{x^2+4} - \frac{1}{x+2}$$

$$\begin{aligned}
 \text{(ii)} \int \frac{5x+2}{(x^2+4)(x+2)} dx &= \int \frac{x+3}{x^2+4} - \frac{1}{x+2} dx \\
 &= \int \frac{x}{x^2+4} + \frac{3}{x^2+4} - \frac{1}{x+2} dx \\
 &= \frac{1}{2} \int \frac{2x}{x^2+4} dx + 3 \int \frac{dx}{x^2+4} - \int \frac{dx}{x+2} \\
 &= \frac{1}{2} \ln(x^2+4) + \frac{3}{2} \tan^{-1}\left(\frac{x}{2}\right) - \ln|x+2| + C
 \end{aligned}$$

Q3

(a) (i) $xy = 9$ $c^2 = 9$ $c = 3$ $\frac{1}{2}a^2 = 9$ $a = 3\sqrt{2}$ [C.R.L.]

$OA = \sqrt{3^2 + 3^2} = 3\sqrt{2}$ $A(1,1) \rightarrow A(3,3)$
 given $e = \sqrt{2}$ or $OS = \frac{\sqrt{(3\sqrt{2})^2 + (3\sqrt{2})^2}}{2}$

$OS = ae$
 $= 3\sqrt{2} \times \sqrt{2}$
 $= 6$

(ii) Q is reflection of P across $y=x$
 $Q\left(\frac{3}{x_0}, 3x_0\right)$

or $d = \frac{|1 \cdot 3x_0 + 1 \cdot \frac{3}{x_0} - 1|}{\sqrt{1^2+1^2}}$

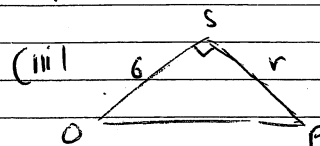
Distance formula $2r = \sqrt{\left(3x_0 - \frac{3}{x_0}\right)^2 + \left(\frac{3}{x_0} - 3x_0\right)^2}$

$\downarrow \left(3x_0 - \frac{3}{x_0}\right)^2 = \left(\frac{3}{x_0} - 3x_0\right)^2$ $(2r)^2 = 2\left(3x_0 - \frac{3}{x_0}\right)^2$

$r = \frac{3x_0 - \frac{3}{x_0}}{\sqrt{2}}$ $\therefore 2r^2 = \left(3x_0 - \frac{3}{x_0}\right)^2$

$r^2 = \left(x_0 - \frac{1}{x_0}\right)^2$

$2r^2 = \left(x_0 - \frac{1}{x_0}\right)^2$



(iii) $OP^2 = OS^2 + SP^2$
 (Pythagoras)
 $OP^2 = 6^2 + r^2$

$OP^2 = 36 + \frac{1}{2} \left(3x_0 - \frac{3}{x_0}\right)^2$ (1)

$\therefore OP^2 = (3x_0 - 0)^2 + \left(\frac{3}{x_0} - 0\right)^2$
 $= \left(3x_0\right)^2 + \left(\frac{3}{x_0}\right)^2$

now $\left(3x_0 - \frac{3}{x_0}\right)^2 = \left(3x_0\right)^2 - 18 + \left(\frac{3}{x_0}\right)^2$

$\therefore \left(3x_0\right)^2 + \left(\frac{3}{x_0}\right)^2 = \left(3x_0 - \frac{3}{x_0}\right)^2 + 18$

From (i) $= 2r^2 + 18$
 $\therefore OP^2 = 2r^2 + 18$
 $= 2(r^2 + 9)$

(iv) QP is latus rectum [focal chord // to directrix
b to axis]

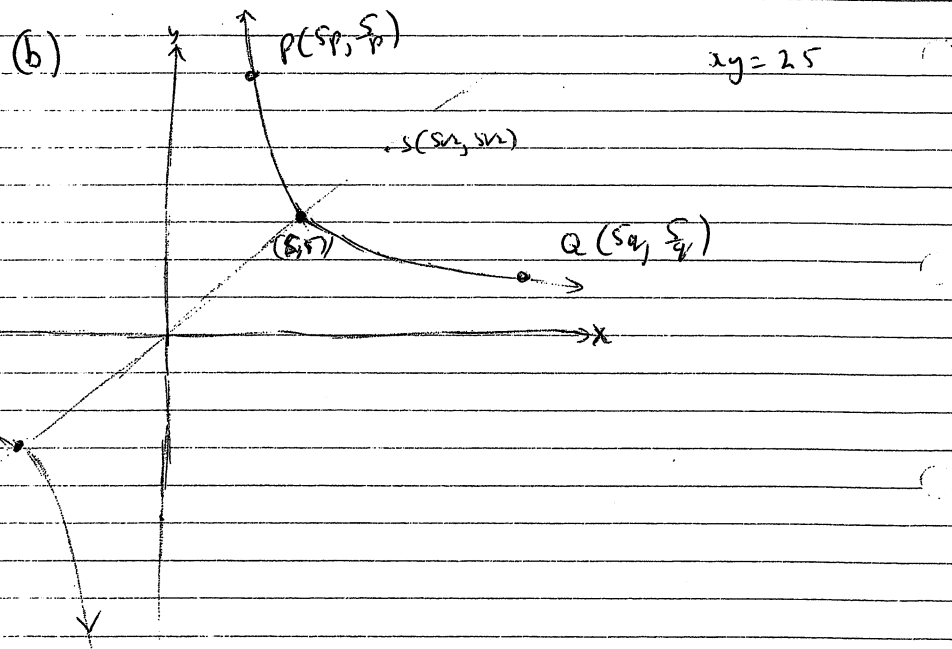
from (iii) $2(q+r^2) = 6+r^2$, [OP =]

$$18+2r^2 = 36+r^2$$

$$r^2 = 18$$

$$r = 3\sqrt{2}$$

∴ length of latus rectum $2r = 6\sqrt{2}$



(i) $x = 5p$, $y = 5p$
 $\frac{dx}{dp} = 5$, $\frac{dy}{dp} = \frac{5}{p^2}$

∴ $\frac{dy}{dx} = -\frac{1}{p^2}$ |

OR $\frac{dy}{dx} = -\frac{2.5}{x^2}$

$x = 5p \rightarrow m = -\frac{1}{p^2}$

Equation of tangent
 $y - \frac{5}{p} = -\frac{1}{p^2}(x - 5p)$

$p^2 y - 5p = -x + 5p$

| $x + p^2 y = 10p$ *

Tangent at $P(5p, 5p) \Rightarrow x + p^2 y = 10p$ -- A
 Similarly, Tangent at $Q(5q, 5q) \Rightarrow x + q^2 y = 10q$ -- B

Solve simultaneously to find T |

(A) - (B) $\Rightarrow (p^2 - q^2)y = 10(p - q)$ & $q^2(A) - p^2(B) \Rightarrow (q^2 - p^2)x = 10pq(q - p)$

$y = \frac{10}{p+q}$

$x = \frac{10pq}{p+q}$

T $\left[\frac{10pq}{p+q}, \frac{10}{p+q} \right]$ |

(ii) Equation of PQ • gradient $m = \frac{\frac{5}{p} - \frac{5}{q}}{5p - 5q}$
 $= \frac{5q - 5p}{pq(5p - 5q)}$
 $= -\frac{1}{pq}$

The $y - \frac{5}{p} = -\frac{1}{pq}(x - 5p)$

$pqy - 5q = -x + 5p$ |

$x + pqy = 5(p+q)$ *

(iii) Sub stn (i) into * (ii)

$0 - 2pq = 5(p+q)$

$\frac{pq}{p+q} = -\frac{5}{2}$ |

OR $\frac{p+q}{pq} = -\frac{2}{5}$

P.T.O.

$$X = \frac{10pq}{pq} ; y = \frac{10}{pq}$$

$$\text{the } X = 10x - \frac{5}{2}$$

$$x = -25$$

T moves on the vertical line $x = -25$

Now ~~the~~

• to pass through $(0, -2)$. P and Q must be in 3rd quadrant since gradient PQ ; $m = \frac{-1}{pq}$ is always -ve.

$$\begin{aligned} \text{• When } x = -25, \quad 2y &= 25 \\ -25y &= 25 \end{aligned}$$

$$y = -1$$

• tangent can not intersect inside hyperbola, so $y > -1$

Since $p, q, q < 0$ the $\frac{10}{pq} < 0$
(3rd quadrant).

So locus is line $x = -25$
where $-1 < y < 0$.